

Initially,

$$\frac{V_1}{I_p} = \frac{R}{(N_2/N_1)^2}$$

If the number of turns in the primary is doubled, then

$$\frac{V_1}{I_{p,new}} = \frac{R}{(N_2/2N_1)^2} = 4 \frac{R}{(N_2/N_1)^2} = 4 \frac{V_1}{I_p}$$

$$\Rightarrow I_{p,new} = \frac{I_p}{4} \quad \text{so the answer is option D.}$$

### Q26 Alternative Explanation:

When the number of turns in the primary is doubled, then  $V_s$  will be halved.

Since the load resistor is the same, from  $V_s = I_s R$ , the  $I_s$  will also be halved.

Therefore, since  $\frac{I_{p,f}}{I_{s,f}} = \frac{1}{2} \frac{I_{p,i}}{I_{s,i}}$  due to the change in turns ratio, and  $I_{s,f}$  is half of  $I_{s,i}$ , the new  $I_{p,f}$  is  $\frac{1}{4}$  of the original  $I_{p,i}$ .

